ANDHRA UNIVERSITY COLLEGE OF ENGINEERING (A)

DEPARTMENT OF COMPUTER SCIENCE & SYSTEMS ENGINEERING

AI & MACHINE INTELLIGENCE WING

★ SOLVED MID & SEMESTER EXAMINATION PAPERS ★

AI & Data Engineering Mid & Semester Exam PYQs

Course Code:	AU-AIML-2201	**Academic Scheme:**	II B.Tech II Semester (AU Scheme)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Paper Section A: Artificial Intelligence Agents & Search

Rational agents, Environment types, Uninformed search (BFS, DFS, Uniform Cost), Informed search (A* search, Heuristics admissibility), Adversarial search (Minimax, Alpha-Beta pruning).

Paper Section B: Supervised Machine Learning Algorithms

k-Nearest Neighbors (k-NN), Naive Bayes classifier, Support Vector Machines (SVM), Hyperplanes, Soft margin and Kernel trick.

Paper Section C: Unsupervised Learning & Clustering

k-Means clustering algorithm, Elbow method for optimal k, Hierarchical clustering, Density-based spatial clustering (DBSCAN).

Paper Section D: Neural Networks & Deep Learning Foundations

Perceptron model, Activation functions (Sigmoid, ReLU, Softmax), Multi-layer Perceptron (MLP), Backpropagation algorithm.

Paper Section E: Model Evaluation & Regularization

Bias-Variance tradeoff, Overfitting prevention, L1 (Lasso) and L2 (Ridge) regularization, Cross-validation techniques.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Question Paper Duration	3 Hours
Max Marks	70 Marks
University Examination Division	Andhra University, Visakhapatnam

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